Bulgaria: Public Sector Debt Sustainability Analysis (DSA) – Baseline Scenario

(In percent of GDP, unless otherwise indicated)

Debt, Economic and Market Indicators [1/]

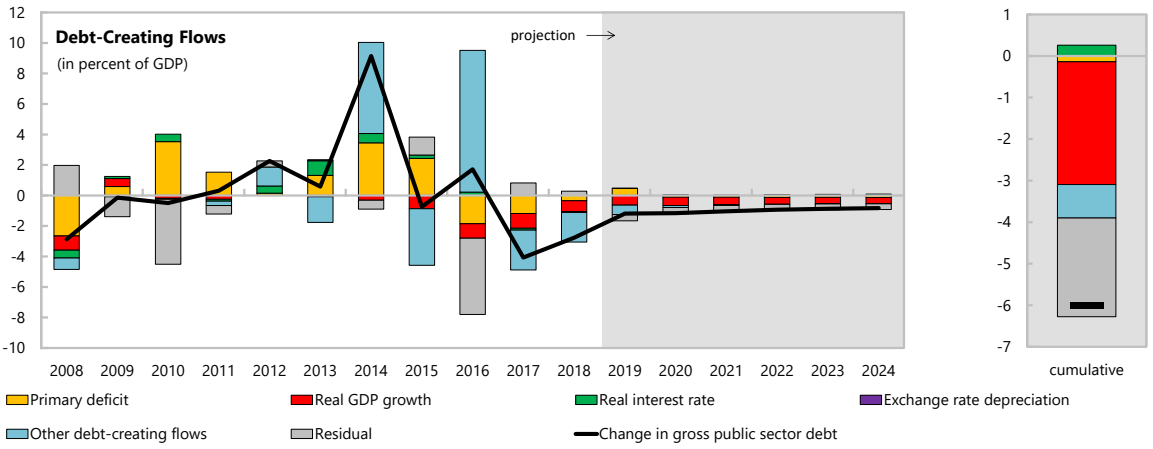
	Actual			Projections					
	2008-2016 [2/]	2017	2018	2019	2020	2021	2022	2023	2024
Nominal gross public debt	19.0	23.3	20.5	19.3	18.2	17.1	16.2	15.3	14.5
Public gross financing needs	4.0	2.1	1.3	1.7	1.2	0.8	2.2	2.1	2.3
Real GDP growth (in percent)	1.7	3.8	3.2	3.3	3.0	2.8	2.8	2.8	2.8
Inflation (GDP deflator, in percent)	2.8	3.4	3.1	2.9	2.8	2.8	2.8	2.8	2.8
Nominal GDP growth (in percent)	4.6	7.3	6.4	6.3	5.8	5.7	5.7	5.7	5.7
Effective interest rate (in percent) [4/]	4.5	3.1	2.9	3.0	3.0	3.0	3.1	3.3	3.5

As of February 21, 2019

Sovereign Spreads		
EMBIG (bp) 3/		149
5Y CDS (bp)		79
Ratings	Foreign	Local
Moody's	Baa2	Baa2
S&Ps	BBB-	BBB-
Fitch	BBB	BBB

Contribution to Changes in Public Debt

	Actual			Projections							
	2008-2016	2017	2018	2019	2020	2021	2022	2023	2024	cumulative	debt-stabilizing primary balance [9/]
Change in gross public sector debt	1.1	-4.1	-2.8	-1.2	-1.2	-1.0	-0.9	-0.9	-0.8	-6.0	
Identified debt-creating flows	2.0	-4.9	-3.1	-0.8	-0.8	-0.6	-0.5	-0.5	-0.5	-3.6	
Primary deficit	0.9	-1.2	-0.3	0.5	-0.1	-0.1	-0.1	-0.1	-0.1	-0.1	-0.3
Primary (noninterest) revenue and grants	32.8	32.9	34.8	36.0	35.3	35.0	35.1	35.1	34.3	210.8	
Primary (noninterest) expenditure	33.8	31.8	34.4	36.5	35.2	34.8	35.0	35.0	34.1	210.7	
Automatic debt dynamics [5/]	-0.1	-1.1	-0.8	-0.6	-0.5	-0.5	-0.4	-0.4	-0.3	-2.7	
Interest rate/growth differential [6/]	-0.1	-1.1	-0.8	-0.6	-0.5	-0.5	-0.4	-0.4	-0.3	-2.7	
Of which: real interest rate	0.3	-0.1	-0.1	0.0	0.0	0.0	0.0	0.1	0.1	0.3	
Of which: real GDP growth	-0.3	-1.0	-0.7	-0.6	-0.5	-0.5	-0.5	-0.4	-0.4	-3.0	
Exchange rate depreciation [7/]	0.0	0.0	0.0	...	...	...	...	...	...	...	
Other identified debt-creating flows	1.1	-2.6	-1.9	-0.6	-0.1	-0.1	0.0	0.0	0.0	-0.8	
0 (negative)	-0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Contingent liabilities	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Please specify (2) (e.g., ESM and Euroarea loans)	1.3	-2.6	-1.9	-0.6	-0.1	0.0	0.0	0.0	0.0	-0.8	
Residual, including asset changes [8/]	-0.9	0.8	0.3	-0.4	-0.4	-0.4	-0.4	-0.4	-0.4	-2.4	



Source: IMF staff.

1/ Public sector is defined as general government.

2/ Based on available data.

3/ Long-term bond spread over German bonds.

4/ Defined as interest payments divided by debt stock (excluding guarantees) at the end of previous year.

5/ Derived as [(r - π(1+g) - g + ae(1+r))/(1+g+π+gπ)) times previous period debt ratio, with r = interest rate; π = growth rate of GDP deflator; g = real GDP growth rate; a = share of foreign-currency denominated debt; and e = nominal exchange rate depreciation (measured by increase in local currency value of U.S. dollar).

6/ The real interest rate contribution is derived from the numerator in footnote 5 as r - π (1+g) and the real growth contribution as -g.

7/ The exchange rate contribution is derived from the numerator in footnote 5 as ae(1+r).

8/ Includes asset changes and interest revenues (if any). For projections, includes exchange rate changes during the projection period.

9/ Assumes that key variables (real GDP growth, real interest rate, and other identified debt-creating flows) remain at the level of the last projection year.

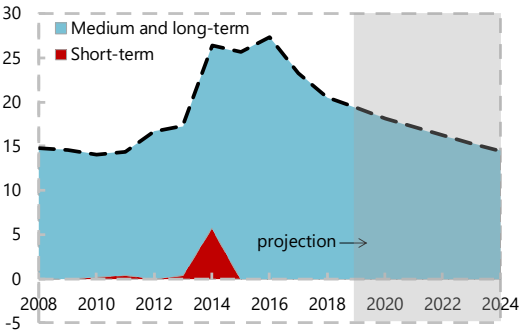
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Bulgaria: Public DSA – Composition of Public Debt and Alternative Scenarios

Composition of Public Debt

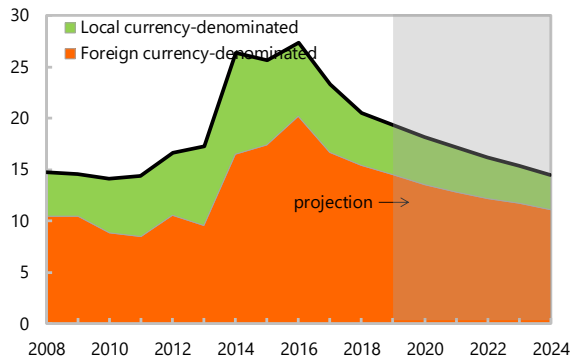
By Maturity

(in percent of GDP)



By Currency

(in percent of GDP)

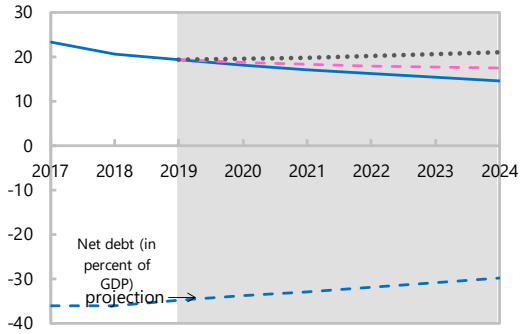


Alternative Scenarios

Baseline — Historical — Constant Primary Balance

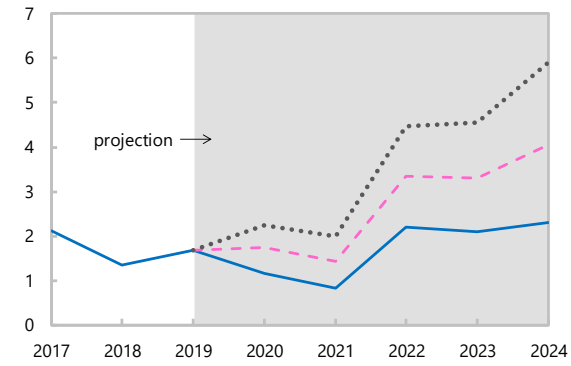
Gross Nominal Public Debt

(in percent of GDP)



Public Gross Financing Needs

(in percent of GDP)



Underlying Assumptions

(in percent)

Baseline Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	3.3	3.0	2.8	2.8	2.8	2.8
Inflation	2.9	2.8	2.8	2.8	2.8	2.8
Primary Balance	-0.5	0.1	0.1	0.1	0.1	0.1
Effective interest rate	3.0	3.0	3.0	3.1	3.3	3.5
Constant Primary Balance Scenario						
Real GDP growth	3.3	3.0	2.8	2.8	2.8	2.8
Inflation	2.9	2.8	2.8	2.8	2.8	2.8
Primary Balance	-0.5	-0.5	-0.5	-0.5	-0.5	-0.5
Effective interest rate	3.0	3.0	3.0	3.2	3.4	3.7

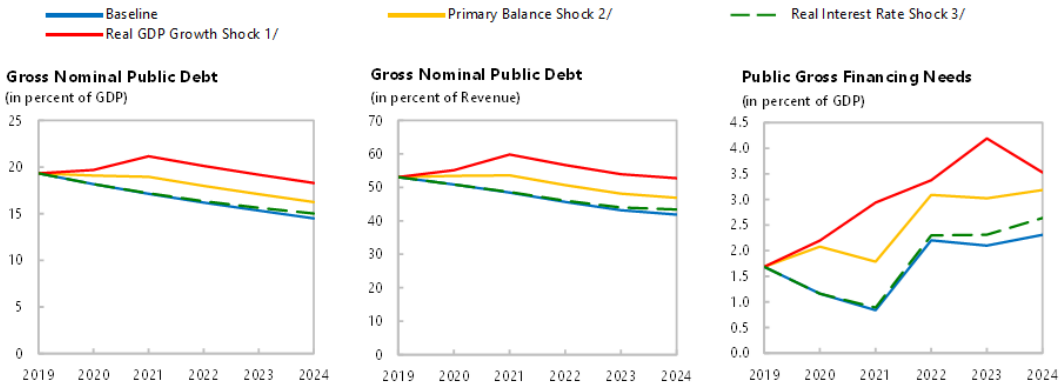
Historical Scenario	2019	2020	2021	2022	2023	2024
Real GDP growth	3.3	1.6	1.6	1.6	1.6	1.6
Inflation	2.9	2.8	2.8	2.8	2.8	2.8
Primary Balance	-0.5	-1.0	-1.0	-1.0	-1.0	-1.0
Effective interest rate	3.0	3.0	3.1	3.4	3.9	4.4

Source: IMF staff.

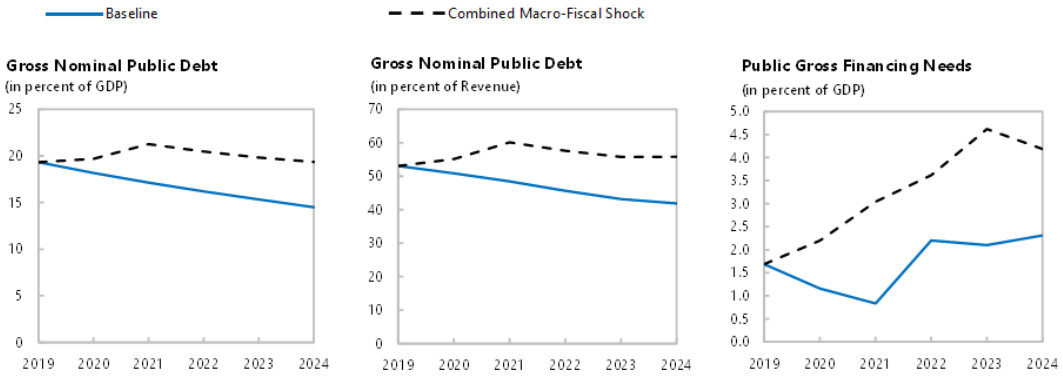
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Bulgaria: Public DSA – Stress Tests

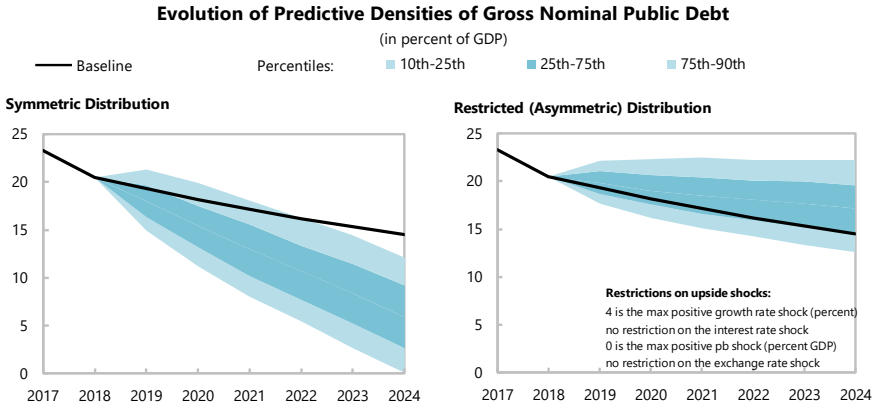
Macro-Fiscal Stress Tests



Additional Stress Tests



Evolution of Predictive Densities of Gross Nominal Public Debt



Source: IMF staff.

1/ Real GDP growth shock scenario assumes (i) real GDP growth is reduced by 1 standard deviation for 2020-21; (ii) revenue-to-GDP ratio remains the same as in the baseline; (iii) level of non-interest expenditures is the same as in the baseline; (iv) deterioration in primary balance lead to higher interest rate; and (v) decline in growth leads to lower inflation (0.25 percentage points per 1 percentage point decrease in GDP growth).

2/ Primary balance shock scenario assumes (i) the primary balance for 2020-21 to be below the baseline projections by 50 percent of the 10-year historical standard deviation; and (ii) additional borrowing leads to increase in interest rate of 25bp per 1 percent of GDP worsening of the primary balance.

3/ Real interest rate shock scenario assumes nominal interest rate increases by the difference between the maximum real interest rate over history (last 10 years) and the average real interest rate level over projection.

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